Congenital Myopathy 28 With Rigid Spine (CMYO28) — Comprehensive Disease Report

MONDO: MONDO:0980756 · **OMIM phenotype:** 621433 · **Causal gene: HMGCS1** (3-hydroxy-3-methylglutaryl-CoA synthase 1; cytosolic) · **Category:** Mendelian, autosomal recessive

> **Causal gene stated up front:** This report concerns the disease caused by biallelic variants in **HMGCS1** (HGNC:5007; NCBI Gene 3157; chromosome 5p12; gene alias **CMYO28**), the first committed enzyme of the cytosolic **mevalonate pathway**. It is *not* the SELENON-related rigid spine muscular dystrophy (RSMD1), with which it overlaps clinically. The near-totality of disease-specific knowledge derives from a single defining study: **Dofash et al., *Brain* 2025;148(5):1707–1722** (P 39531736).
>
> Where a section has no disease-specific evidence, this is stated explicitly and inferences are drawn from mevalonate-pathway biology and the closely related HMGCR/GGPS1 myopathies.

1. Disease Information

Overview. Congenital Myopathy 28 with Rigid Spine (CMYO28) is a rare, childhood-onset recessive myopathy belonging to the "rigid spine syndrome" clinical group. It is characterized by slowly progressive or non-progressive **spinal rigidity (cervical and dorso-lumbar), scoliosis, axial/limb contractures, hypotonia, and respiratory insufficiency**, with variably elevated creatine kinase (CK). It was defined in 2025 as the newest recognized mevalonate-pathway myopathy. Rigid spine syndrome as a group is "a rare childhood-onset myopathy characterized by slowly progressive or non-progressive scoliosis, neck and spine contractures, hypotonia and respiratory insufficiency"

(P 39531736);
most such cases are caused by SELENON, and HMGCS1 explains a subset of previously unsolved cases.

Key identifiers. - **OMIM phenotype:** 621433 (Congenital Myopathy 28 With Rigid Spine, CMYO28) - **Gene OMIM:** HMGCS1 — 142940 - **MONDO:** MONDO:0980756 - **HGNC:** HMGCS1, HGNC:5007; **NCBI Gene:** 3157; **Ensembl:** ENSG00000112972; **UniProt:** Q01581 (HMCS1_HUMAN) - **ICD-11:** best mapped to 8C70 (Primary disorders of muscles / congenital myopathies); **ICD-10:** G71.2 (congenital myopathies) — no CMYO28-specific code exists. - **MeSH:** no specific descriptor; closest umbrella terms are "Myopathies, Structural, Congenital" (D020914) and "Muscular Diseases." - **Orphanet:** no dedicated ORPHA code yet (disease defined 2024–2025); phenotypically within the "rigid spine syndrome" / congenital myopathy groups.

Synonyms / alternative names. CMYO28; HMGCS1-related rigid spine myopathy; HMGCS1-related rigid spine syndrome; mevalonate-pathway rigid spine myopathy. Gene aliases: HMGCS, HMG-CoA synthase 1 (soluble/cytosolic).

Information source type. Aggregated from **individual patient reports** (5 patients from 4 unrelated families) combined with in vitro recombinant-protein biochemistry and a zebrafish disease model — i.e., primary-literature case-level data, not population registries or EHR aggregation.

---

2. Etiology

Primary cause (genetic). Biallelic (autosomal recessive) variants in **HMGCS1**. "In five patients from four unrelated families, we identified biallelic variants in HMGCS1 ... six missense variants and one frameshift variant distributed throughout HMGCS1" (P 39531736).
HMGCS1 catalyzes condensation of acetyl-CoA + acetoacetyl-CoA → HMG-CoA, the first committed, cytosolic step of the mevalonate pathway that produces cholesterol and non-sterol isoprenoids (dolichol, ubiquinone/CoQ10, farnesyl- and geranylgeranyl-pyrophosphate for protein prenylation).

Genetic risk factors. The disease is monogenic/Mendelian: the only established risk factor is inheriting **two** pathogenic/hypomorphic HMGCS1 alleles. Heterozygous carriers are unaffected. No susceptibility loci or modifier genes have been formally identified.

Mechanistically, variants in **downstream** mevalonate-pathway enzymes act as candidate phenocopy/comparator loci: "biallelic hypomorphic variants in downstream enzymes including HMGCR and GGPS1 are associated with muscular dystrophy resembling our cohort's presentation"
(39531736).

Environmental / acquired modifiers. Not causal, but clinically important as **triggers of deterioration**: "The clinical course worsened with intercurrent disease or certain drugs in some patients; one patient died from respiratory failure following infection"
(39531736).

By mechanistic inference, **statins** (HMG-CoA reductase inhibitors) and other mevalonate-pathway–suppressing drugs are expected to be deleterious, since statins alone cause myopathy by inhibiting the immediately downstream enzyme HMGCR
(36745799).
37167966

Age, sex, occupational, and lifestyle risk factors are not established.

Protective factors. No genetic protective alleles described. Mechanistically, **mevalonic acid supplementation** is protective in the zebrafish model (see §12) — bypassing the deficient enzymatic step restores downstream flux.

Gene–environment interaction. Strongly suggested: a partially functional (hypomorphic) mevalonate pathway is **decompensated by metabolic stress** (infection, fever, catabolic illness) and by pathway-inhibiting drugs, precipitating respiratory crises. This is an inferred gene × environment interaction based on the clinical course description, not a formally quantified interaction.

---

3. Phenotypes

Frequencies below are the **curated HPO disease annotations for MONDO:0980756** (Monarch/HPOA, derived from the

P 39531736

cohort), which give explicit percentages. Phenotype type and HPO terms are given; obligate (100%) features are the diagnostic core.

Phenotype	Type	Frequency	Characteristics	HPO term
Spinal rigidity (cervical + dorso-lumbar)	Clinical sign / contracture	**100%**	Childhood onset; slowly/non-progressive; defining feature	**HP:0003306** (Spinal rigidity)
Scoliosis	Physical manifestation	**100%**	Childhood onset; may progress	**HP:0002650** (Scoliosis)
Elevated creatine kinase	Laboratory abnormality	**100%** (variably elevated level)	"Creatine kinase levels were variably elevated"	**HP:0003236** (Elevated circulating creatine kinase activity)
Fatty replacement of skeletal muscle	Imaging/ pathology	**100%**	Muscle MRI/biopsy fatty infiltration	**HP:0012548** (Fatty replacement of skeletal muscle)
Reduced forced vital capacity / respiratory insufficiency	Functional / clinical sign	**100%** (reduced FVC)	Restrictive; diaphragmatic weakness; cause of death in 1 patient	**HP:0032341** (Reduced forced vital capacity); HP:0002093 (Respiratory insufficiency)
Rimmed vacuoles (muscle biopsy)	Pathology	**100%**	Plus oxidative-enzyme-stain irregularities, occasional internal nuclei	**HP:0003805** (Rimmed vacuoles); HP:0003687 (Centrally nucleated fibers)
Scapular winging	Clinical sign	**80%**	Axial/scapular girdle involvement	**HP:0003691** (Scapular winging)
Proximal muscle weakness	Clinical sign	**80%**	Limb-girdle component	**HP:0003701** (Proximal muscle weakness)
	Clinical sign	**60%**		

Phenotype	Type	Frequency	Characteristics	HPO term
Distal muscle weakness				**HP:0002460** (Distal muscle weakness)
High/narrow palate	Physical manifestation	**40%**	Dysmorphic/ secondary to weakness	**HP:0000218** (High palate)
Bulbar palsy	Clinical sign	**20%**	Bulbar weakness	**HP:0001283** (Bulbar palsy)
Hypotonia	Clinical sign	characteristic of group (congenital)	Often early	**HP:0001252** (Hypotonia)
Clinical worsening with illness/drugs; risk of death	Course modifier	some	Respiratory decompensation; 1 death	HP:0002878 (Respiratory failure)

Explicitly absent / discriminating negatives (0% in annotations): facial muscle weakness (HP:0030319), ptosis (HP:0000508), **reduced left-ventricular ejection fraction (HP:0012664) — i.e., no primary cardiomyopathy**, and myositis (HP:0100614). Sparing of face, eyelids and heart helps distinguish CMYO28 from other congenital myopathies.

Onset: childhood-onset (congenital myopathy group); **severity:** moderate–severe and **variable**; **progression:** slowly progressive or non-progressive baseline with **episodic decompensation** during intercurrent illness.

Quality-of-life impact. Not directly measured in the HMGCS1 cohort. By close analogy to SELENON-related rigid spine myopathy, expect impaired QoL with pain, problematic fatigue, restricted mobility/standing-transfers, and respiratory burden: SELENON-RM "Questionnaires revealed impaired quality of life, pain and problematic fatigue" and diaphragm/respiratory function "was impaired in all patients" (P 37807786). (Evidence: analogous disease, not HMGCS1-specific.)

4. Genetic / Molecular Information

Causal gene. HMGCS1 (OMIM 142940; HGNC:5007; Gene 3157), 12 exons on 5p12 (GRCh38 NC_000005.10:~43,287,469–43,313,411, minus strand). Protein: **cytoplasmic hydroxymethylglutaryl-CoA synthase**, **520 aa**, **EC 2.3.3.10**, functions as a **homodimer**; UniProt **Q01581**; crystal structure **PDB 2P8U**; domains Pfam **PF01154** (HMG_CoA_synt_N) + **PF08540** (HMG_CoA_synt_C), InterPro IPR013528/IPR013746/IPR010122 (thiolase-like fold IPR016039); subcellular location **cytoplasm** (UniProt). Canonical transcript NM_001098272 / NM_002130. Function (UniProt): "Catalyzes the condensation of acetyl-CoA with acetoacetyl-CoA to form HMG-CoA, which is converted by HMG-CoA reductase (HMGCR) into mevalonate, a precursor for cholesterol synthesis."

Pathogenic variants (from

P 39531736

;

HGVS confirmed against ClinVar, transcript NM_001098272). - **Seven variants total:** six missense + one frameshift, distributed throughout the gene (no single hotspot). - **Specific alleles (HGVS): c.209T>C (p.Met70Thr)**, **c.803G>C (p.Cys268Ser)**, **c.890G>T (p.Gly297Val)**, **c.1289G>A (p.Arg430Lys)**, **p.Ser447Pro**, **p.Gln29Leu** (six missense), and **c.344_345del (p.Ser115fs)** (frameshift). In ClinVar, p.Gly297Val, p.Arg430Lys and p.Ser115fs are classified **Pathogenic**; p.Met70Thr and p.Cys268Ser show "conflicting classifications" (reflecting their recent, 2025 disease association). - **Functionally characterized missense variants: p.Ser447Pro, p.Gln29Leu, p.Met70Thr, p.Cys268Ser** (recombinant-protein and zebrafish-rescue assays). - **Variant type/class:** predominantly **missense**; one **frameshift** (predicted truncating/LoF). - **Functional consequence: hypomorphic / partial loss of function** — "these missense variants in HMGCS1 act through a hypomorphic mechanism." All four tested mutants retained dimerization; 3/4 had reduced thermal stability; 2/4 showed subtle enzymatic-activity changes; all four showed reduced function in zebrafish rescue. Biallelic complete LoF is presumed non-viable (HMGCS1 is essential; see zebrafish lethality), so disease genotypes are hypomorphic. - **ACMG/AMP classification:** reported as disease-causing with functional support (PS3/PM-level evidence); several alleles curated **Pathogenic** in ClinVar, others still "conflicting/VUS" pending wider recognition. - **Allele frequency / population constraint:** disease alleles are rare/very low frequency in gnomAD. **HMGCS1 is extremely loss-of-function intolerant: gnomAD pLI = 0.99999, LOEUF (oe_lof upper 90% CI) = 0.30 (point o/e 0.17), LoF Z = 5.08, missense Z = 5.15** — strong population depletion of both LoF and missense

alleles, consistent with essentiality and with viable disease requiring residual-function (hypomorphic) genotypes. - **Origin: germline**, biallelic. No somatic involvement (not a neoplastic disease).

Modifier genes. None established. Downstream pathway load (HMGCR, GGPS1, FDPS, etc.) is a plausible modifier space but untested.

Epigenetics / chromosomal abnormalities. No disease-specific methylation, histone, or large structural/cytogenetic findings reported. Not a copy-number/aneuploidy disorder.

---

5. Environmental Information

- **Environmental/chemical factors:** No causal environmental agent. **Drugs that inhibit the mevalonate pathway (statins; possibly bisphosphonates/aminobisphosphonates acting on FDPS; antifungal/other mevalonate-suppressing agents) are expected to aggravate disease** by mechanistic analogy to statin myopathy and to the reported "certain drugs" worsening the course
  (P 39531736).
  CHEBI terms: statins (CHEBI:87631), mevalonic acid (CHEBI:25350), cholesterol (CHEBI:16113), ubiquinone (CHEBI:16389).
- **Lifestyle factors:** No established diet/smoking/exercise risk. Catabolic stress and dehydration during illness plausibly worsen the metabolic myopathy.
- **Infectious agents:** None causal, but **intercurrent infection is a documented precipitant of fatal respiratory decompensation**
  (P 39531736).

---

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic hypomorphic HMGCS1 variants** (missense ± frameshift) **lead to** reduced thermal stability and modestly reduced catalytic activity of the cytosolic HMG-CoA synthase homodimer (demonstrated in vitro). *[Demonstrated: recombinant protein assays.]*
2. This **results in** decreased synthesis of **HMG-CoA**, the first committed intermediate of the mevalonate pathway. *[Inferred from enzyme function; HMGCS1 catalyzes acetyl-CoA + acetoacetyl-CoA → HMG-CoA.]*
3. Reduced HMG-CoA **leads to** reduced flux to **mevalonate** and thence to all downstream isoprenoids — **cholesterol, dolichol, ubiquinone (CoQ10), and farnesyl-/geranylgeranyl-pyrophosphate (FPP/GGPP)**. *[Inferred from pathway position; supported by rescue with exogenous mevalonic acid.]*
4. **Branch A — defective protein prenylation:** reduced GGPP/FPP **leads to** impaired prenylation of small GTPases (e.g., Rab/Rho), paralleling GGPS1-related muscular dystrophy, where loss of geranylgeranyl-PP causes dystrophic muscle, **delayed sarcolemmal membrane repair** in patient myogenic cells, and enlarged mitochondria/autophagic material
   (P 32403198).
   [Inferred for HMGCS1; demonstrated for the downstream enzyme GGPS1.]
5. **Branch B — membrane/sterol and CoQ deficits:** reduced cholesterol/dolichol/ubiquinone **leads to** membrane and mitochondrial/oxidative dysfunction in muscle, consistent with biopsy "irregularities in oxidative enzyme staining" and rimmed vacuoles. *[Partly demonstrated (biopsy); mechanism inferred.]*
6. These deficits **result in** myofiber dysfunction/degeneration and impaired regeneration in **axial (paraspinal, cervical) and diaphragmatic muscle**, producing **spinal rigidity, scoliosis, contractures, and respiratory insufficiency**. *[Demonstrated clinically/histologically; cell-biological link inferred.]*
7. Under **metabolic stress (infection, catabolic illness) or pathway-inhibiting drugs**, the already marginal mevalonate flux **decompensates**, **leading to** acute respiratory failure and, in one patient, **death**. *[Demonstrated clinically.]*

8. In the zebrafish model, loss of Hmgcs1 **leads to** immobility by 2 dpf and death by day 3; **supplying mevalonic acid bypasses step 2–3 and reduces phenotype severity**, closing the causal loop. *[Demonstrated in animal model.]*

Checklist detail

- **Molecular pathways:** Mevalonate / isoprenoid / cholesterol biosynthesis (KEGG hsa00900 "Terpenoid backbone biosynthesis"; Reactome R-HSA-191273 "Cholesterol biosynthesis"). GO:0019287 (isopentenyl diphosphate biosynthetic process, mevalonate pathway); GO:0010142 (farnesyl diphosphate biosynthetic process); GO:0006695 (cholesterol biosynthetic process).
- **Cellular processes:** oxidative metabolism/mitochondrial function (biopsy oxidative-stain irregularities), autophagic/vacuolar pathology (rimmed vacuoles), impaired muscle regeneration (inferred).
- **Protein dysfunction:** hypomorphic HMGCS1 — reduced thermostability, subtle activity loss, **dimerization preserved** (not a dominant-negative/aggregation mechanism). Enzyme activity: GO:0004421 (hydroxymethylglutaryl-CoA synthase activity).
- **Metabolic changes:** reduced endogenous cholesterol and non-sterol isoprenoid (dolichol, CoQ, prenyl-PP) synthesis in muscle.
- **Immune involvement:** none established (distinct from anti-HMGCR *immune-mediated* necrotizing myopathy, which is autoantibody-driven and unrelated).
- **Tissue damage:** myofiber degeneration with internal nuclei, rimmed vacuoles, oxidative-enzyme irregularity.
- **Biochemical abnormality:** partial HMG-CoA synthase deficiency (EC 2.3.3.10).
- **Molecular profiling:** no transcriptomic/proteomic/metabolomic patient datasets published yet; proteomics expertise (Stroud lab) contributed to functional work. **Functional genomics/model:** zebrafish *hmgcs1* mutant (lethal, mevalonate-rescuable).
- **Cell types:** skeletal myofibers (**CL:0000188** skeletal muscle cell), satellite cells (**CL:0000594**) plausibly involved in impaired regeneration.

---

7. Anatomical Structures Affected

- **Primary organ/system: skeletal muscle**, musculoskeletal and respiratory systems. Axial muscles — paraspinal/cervical (**UBERON:0001134** skeletal muscle tissue; UBERON:0001774 neck muscle) — and the **diaphragm (UBERON:0001103)**.
- **Secondary involvement:** spine/vertebral column (**UBERON:0001130**) → scoliosis and rigidity; **lungs (UBERON:0002048)** / respiratory function secondarily (restrictive respiratory failure); possible secondary **right-heart/pulmonary-hypertension** burden (reported in the analogous SELENON-RM group,

  P 35868898

  — not confirmed in HMGCS1).
- **Tissue level:** striated muscle tissue; myopathic changes on biopsy.
- **Cell level:** skeletal myofibers (CL:0000188); satellite/muscle stem cells (CL:0000594) by inference.
- **Subcellular (GO Cellular Component): cytosol (GO:0005829)** — site of HMGCS1 and the mevalonate pathway; secondary **mitochondrial (GO:0005739)** oxidative-enzyme abnormalities; **endoplasmic reticulum (GO:0005783)** (downstream cholesterol/dolichol synthesis); autophagic **vacuoles** (rimmed vacuoles).
- **Localization/lateralization: bilateral, axial-predominant** (cervical and dorso-lumbar spine, diaphragm); scoliosis introduces a lateralized curvature but the muscle disease is symmetric.

---

8. Temporal Development

- **Onset: childhood-onset / congenital myopathy** spectrum; rigid spine syndrome is "childhood-onset"
  (P 39531736).
  Onset pattern **insidious/chronic** with early hypotonia.
- **Progression: slowly progressive or non-progressive** scoliosis/rigidity baseline, with superimposed **episodic acute decompensation** during intercurrent illness or drug exposure.

- **Course:** chronic, lifelong; respiratory function tends to decline; **critical vulnerability windows = intercurrent infections and exposure to mevalonate-inhibiting drugs.**
- **Remission:** no spontaneous remission; mevalonic acid supplementation is a candidate disease-modifying (not curative) intervention (animal data only).
- **Critical period for intervention:** restoring mevalonate flux (supplementation) and aggressive respiratory/infection management during catabolic illness.

---

9. Inheritance and Population

- **Inheritance: autosomal recessive** (biallelic). Consistent across 4 unrelated families.
- **Penetrance/expressivity:** apparently high penetrance for biallelic genotypes; **variable expressivity** (CK variable; severity ranges to early death). Formal penetrance estimates not available (n=5).
- **Carrier frequency / founder effects:** not established; individual disease variants are rare/very low frequency in gnomAD, and HMGCS1 is strongly LoF- and missense-constrained (pLI≈1.0, LOEUF 0.30), so biallelic null carriers are essentially absent. No founder haplotype reported (families unrelated, geographically diverse — Australia, Spain, Italy, France, Japan, USA contributing centers).
- **Epidemiology: ultra-rare**; only 5 patients reported worldwide at description. No prevalence/incidence estimates; no dedicated registry.
- **Demographics:** no sex predilection established (recessive, small n); multi-ethnic (European, Japanese, and others represented through contributing centers). Age distribution: pediatric onset, into adulthood.
- **Anticipation / germline mosaicism:** not applicable / not reported (not a repeat-expansion disorder).
- **Consanguinity:** biallelic recessive; some families may be consanguineous (typical for homozygous hypomorphic genotypes), though specific consanguinity details are not extractable here.

---

10. Diagnostics

- **Laboratory:** serum **creatine kinase** — variably elevated (HP:0003236; LOINC 2157-6, Creatine kinase [serum]). No validated disease-specific circulating biomarker yet; a metabolite signature (low mevalonate/isoprenoid intermediates) is a plausible but unproven biomarker.
- **Muscle biopsy / histopathology:** irregular **oxidative enzyme (NADH/SDH/COX) staining**, occasional **internal (central) nuclei**, and **rimmed vacuoles** — nonspecific myopathic/dystrophic changes
  (P 39531736).
  (SNOMED: myopathy, muscle biopsy abnormal.)
- **Imaging:** spine radiographs/MRI for scoliosis and rigidity; muscle MRI/ultrasound expected to show axial/paraspinal involvement (by analogy to SELENON-RM, which shows symmetric increased echogenicity;
  P 37807786).
  RadLex: scoliosis, muscle atrophy.
- **Functional/respiratory testing:** pulmonary function tests with emphasis on **diaphragm function** (restrictive pattern; nocturnal hypoventilation surveillance), as in SELENON-RM
  (P 37807786).
  Cardiac surveillance (ECG/echo) advisable given secondary pulmonary-hypertension risk in the rigid-spine group
  (P 35868898).
- **Genetic testing (diagnostic gold standard): exome or genome sequencing** identified the variants
  (P 39531736).
  Recommended approach: WES/WGS or a **congenital myopathy / rigid-spine gene panel** that now should **include HMGCS1** alongside SELENON, and consider HMGCR and GGPS1. Single-gene HMGCS1 testing is reasonable when phenotype is classic and panel negative. CMA/karyotype/FISH/mtDNA/repeat-expansion testing are **not** indicated (point-mutation recessive disorder).
- **Clinical criteria / differential diagnosis:** diagnose rigid spine myopathy clinically (early spinal rigidity + scoliosis + respiratory insufficiency + relative preservation of ambulation),

then confirm genetically. **Differentials:** SELENON (SEPN1)-related rigid spine muscular dystrophy (RSMD1) — most common cause; collagen VI–related (Ullrich/Bethlem); LMNA-related; FHL1; multiminicore/RYR1; HMGCR- and GGPS1-related muscular dystrophies (same pathway); and **immune-mediated anti-HMGCR necrotizing myopathy** (serologic, treatable, mechanistically distinct from HMGCS1).

- **Screening:** no newborn screening. **Cascade/carrier testing** in families; prenatal/preimplantation testing possible once biallelic variants are known.

---

11. Outcome / Prognosis

- **Survival/mortality:** potentially fatal — **"one patient died from respiratory failure following infection"** (P 39531736). No cohort survival statistics (n=5).
- **Morbidity/function:** chronic restrictive respiratory impairment, scoliosis, contractures, and reduced mobility drive disability; likely need for respiratory support (NIV) and orthopedic/scoliosis management over time (by analogy to SELENON-RM).
- **Disease course:** chronic progressive with acute decompensations; recovery potential limited without disease-modifying therapy.
- **Prognostic factors:** severity of respiratory/diaphragmatic involvement; frequency of intercurrent illness; exposure to mevalonate-inhibiting drugs; (hypothesized) degree of residual HMGCS1 function by genotype.
- **Prognostic biomarkers:** none validated; CK is nonspecific.

---

12. Treatment

No approved disease-specific therapy exists. Management is currently supportive, with one mechanistically rational candidate therapy from animal data.

- **Candidate disease-modifying therapy — mevalonic acid supplementation:** "we demonstrate the potential for mevalonic acid supplementation to reduce phenotypic severity in mutant zebrafish"

( ). P 39531736

Rationale: exogenous **mevalonate (CHEBI:25350)** bypasses the deficient HMGCS1 step, restoring downstream isoprenoid/sterol synthesis. Human efficacy/safety untested. (NCIT: dietary supplement / metabolic replacement therapy.)

- **Avoid pathway inhibitors: statins and other mevalonate-pathway–suppressing drugs should be avoided** (drugs worsened the course; statins alone cause myopathy via the downstream HMGCR step —

P 36745799

).

P 37167966

Pharmacogenomic-style caution: genotype (biallelic HMGCS1) should flag statin contraindication.

- **Supportive/rehabilitative (standard for rigid-spine congenital myopathy):**
- **Respiratory care:** surveillance PFTs, nocturnal non-invasive ventilation, aggressive management of chest infections, vaccination against respiratory pathogens (NCIT: Respiratory Therapy; Mechanical Ventilation).
- **Orthopedic:** scoliosis monitoring, bracing, and spinal fusion surgery when indicated (NCIT: Spinal Fusion).
- **Physiotherapy/OT:** contracture prevention, stretching, mobility aids; nutrition support during catabolic illness.
- **Advanced therapeutics (gene/RNA/cell therapy):** none in development reported; because the mechanism is hypomorphic LoF, **gene replacement or metabolic bypass** are conceptually attractive future directions.
- **Experimental/clinical trials:** none registered specifically for HMGCS1 disease as of the 2025 description.
- **Combination/personalized strategy:** personalized approach = confirm biallelic HMGCS1 genotype → avoid mevalonate inhibitors → standard respiratory/orthopedic supportive care → consider mevalonate supplementation under research/compassionate framework.

---

13. Prevention

- **Primary prevention:** not applicable to a germline Mendelian disease at the population level; **avoid mevalonate-pathway–inhibiting drugs** in known patients (prevents iatrogenic worsening).
- **Secondary prevention:** early genetic diagnosis via panel/exome in rigid-spine phenotypes; proactive **respiratory surveillance** and prompt infection treatment to prevent fatal decompensation.
- **Tertiary prevention:** NIV, scoliosis surgery, contracture management to prevent complications.
- **Genetic counseling:** autosomal recessive — 25% recurrence risk for carrier couples; offer **carrier/cascade testing** and **prenatal/preimplantation genetic testing** once familial variants are defined (NSGC/ACMG framework).
- **Immunization/public health/environmental:** routine vaccination (influenza, pneumococcus, RSV where applicable) is prudent given infection-triggered deterioration; no vector/sanitation dimension.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** HMGCS1 is **deeply conserved**. Human (NCBI Taxon 9606) gene 3157. Orthologs: mouse *Hmgcs1* (NCBI Gene 208715; Taxon 10090), zebrafish *hmgcs1* (Taxon 7955), Drosophila and yeast (ERG13) homologs exist (mevalonate pathway is ancient/essential).
- **Natural disease in other species:** no naturally occurring HMGCS1 myopathy catalogued in OMIA for companion/livestock species to date. Zebrafish *hmgcs1* loss-of-function (the historical *stachel* mutant) is a known developmental-lethal model.
- **Comparative biology:** the lethality of complete loss across species underscores the essentiality of the mevalonate pathway, explaining why only **hypomorphic** biallelic human genotypes are viable and disease-causing. Evolutionary conservation of the pathway supports cross-species mechanistic translation.
- **Zoonosis/transmission:** not applicable (non-infectious genetic disease).

15. Model Organisms

- **Primary model — zebrafish (*Danio rerio*; Taxon 7955):** *hmgcs1* mutant. Phenotype: **immobile by 2 dpf, dead by day 3 post-fertilisation**; rescued by wild-type **HMGCS1 mRNA**. Used for **variant functional validation** (S447P, Q29L, M70T, C268S all showed reduced rescue vs wild-type) and **therapeutic proof-of-concept** (mevalonic acid reduced severity)
(P 39531736).
- *Recapitulation:* models the essential-gene requirement and the hypomorphic allelic series; demonstrates treatability. *Limitation:* embryonic lethality means it models complete loss / severe reduction rather than the slowly progressive axial myopathy of patients; adult muscle-specific phenotype not captured.
- **In vitro model — recombinant human HMGCS1 protein:** biochemical characterization of dimerization, thermal stability, and enzymatic activity for four variants
(P 39531736).
- **Mouse (*Mus musculus*; Taxon 10090) — *Hmgcs1* (MGI:107592; Gene 208715):** IMPC reports the **homozygous knockout is preweaning-lethal with complete penetrance** (Viability Outcome), confirming essentiality and matching zebrafish lethality and the human gnomAD constraint. **Heterozygous** mice show **abnormal auditory brainstem response (hearing)** and **increased circulating ALT (liver)** — intriguingly echoing the hearing loss and hepatic involvement of the downstream GGPS1 mevalonate-pathway disease. No muscle-specific/conditional CMYO28 mouse model is yet reported; such a model would be the logical next tool to capture the slowly progressive axial myopathy that the lethal global null cannot.
- **Comparator pathway models: HMGCR**
(P 37167966) (P 36745799)
and **GGPS1** (Y259C knock-in mouse;
P 32403198).
- **Resources:** ZFIN (zebrafish *hmgcs1*), MGI/IMPC (mouse *Hmgcs1*, MGI:107592), Alliance of Genome Resources.

Supported vs Refuted Hypotheses

Supported: - H1 — Biallelic HMGCS1 variants cause a recessive rigid-spine congenital myopathy (CMYO28). **Supported** (5 patients/4 families; functional validation). - H2 — Disease variants are **hypomorphic** (partial LoF), not dominant-negative. **Supported** (dimerization retained; reduced stability/activity; reduced zebrafish rescue). - H3 — The defect lies in the **mevalonate/isoprenoid pathway**, shared with HMGCR/GGPS1 myopathies. **Supported** (pathway position; phenotypic convergence). - H4 — **Mevalonic acid** can mitigate the phenotype. **Supported in animal model** (zebrafish); unproven in humans. - H5 — HMGCS1 is **developmentally essential**, so only partial-LoF (hypomorphic) alleles are viable and disease-causing; complete biallelic LoF is lethal. **Supported** by convergent cross-species essentiality: zebrafish null death by ~day 3, mouse *Hmgcs1* homozygous preweaning lethality (MGI:107592, IMPC), and extreme human constraint (gnomAD pLI=0.99999, LOEUF=0.30, lof_z=5.08). - H6 — The clinical signature is **axial/rigid-spine–predominant with high-penetrance fatty muscle replacement and restrictive respiratory decline**. **Supported** by Monarch/HPO frequencies: spinal rigidity, scoliosis, elevated CK, fatty muscle replacement, reduced FVC, and rimmed vacuoles each at 100% of reported cases; facial weakness, ptosis, and reduced LVEF absent.

Refuted / not supported: - Dominant-negative or protein-aggregation mechanism — **refuted** (dimerization preserved; recessive inheritance). - Autoimmune mechanism (as in anti-HMGCR myopathy) — **not applicable** to HMGCS1 genetic disease.

Limitations and Future Directions

- Evidence rests essentially on **one cohort (n=5)** and model systems; prevalence, penetrance, natural history, sex ratio, and QoL are **not yet quantified**. HPO frequencies are derived from this small case series, so point estimates (e.g., "100%") are precision-limited.
- No human metabolomic/proteomic/muscle-imaging dataset specific to HMGCS1 patients; **no clinical trial** of mevalonic acid.
- **Future:** patient registry and natural-history study; muscle-specific conditional mouse model; metabolite biomarker (mevalonate/isoprenoid panel); formal statin-contraindication guidance; trial of mevalonic acid or downstream isoprenoid supplementation (CoQ10, GGOH).

Key Evidence Sources

- P 39531736 — Dofash et al., *Brain* 2025;148(5):1707–1722 (defining study; patients, variants, functional, zebrafish, treatment).
- P 37167966 — Morales-Rosado et al., biallelic HMGCR → recessive LGMD (pathway comparator).
- P 36745799 — Yogev et al., HMGCR LoF limb-girdle muscular disease / statin-myopathy mechanism.
- P 32403198 — Foley et al., *Ann Neurol* 2020; GGPS1 → muscular dystrophy/hearing loss/ovarian insufficiency; defines isoprenoid/prenylation branch, Y259C knock-in mouse, delayed membrane repair (pathway comparator).
- P 35869884 — Kaiyrzhanov et al., GGPS1 cohort (hearing loss variable, 46%).
- P 38129970 — Altassan et al., GGPS1 phenotype/genotype expansion (hepatic involvement).
- **UniProt Q01581 / PDB 2P8U** — HMGCS1 protein structure and function annotation.
- P 37807786 — Bouman et al., SELENON-RM natural history/QoL (clinical analog).
- P 35868898 — Bouman et al., cardiac involvement in SELENON-RM (secondary PH risk, analog).